

TECHNICAL ASSESSMENT

Coding Project Assignment

PROJECT DETAILS

Project: Backend API Development - Web Engineer

Position: Web Engineer

Company: Our Company

Level: Mid-level

Duration: 8-12 hours

Technologies: JavaScript, HTML/CSS, Node.js, Express.js

Industry: Technology

WHAT TO BUILD

As part of the Web Engineer role at Our Company, you'll be working on a project that aligns with our Technology industry focus. Build a comprehensive application that demonstrates your skills in JavaScript, HTML/CSS, Node.js, Express.js. This project will evaluate your practical development abilities and how you approach real-world challenges in the Technology industry. This project will help us evaluate your practical skills and how you approach real-world development challenges. This project will test your ability to work with modern web technologies and frameworks.

REQUIREMENTS

1. Design and implement RESTful API with proper endpoints
2. Set up database integration (MongoDB, PostgreSQL, or MySQL)
3. Implement user authentication and authorization (JWT)
4. Add input validation and error handling middleware
5. Create CRUD operations for main entities
6. Implement data filtering, sorting, and pagination
7. Add API documentation with Swagger/OpenAPI
8. Write comprehensive unit and integration tests
9. Implement logging and monitoring capabilities
10. Add security measures and rate limiting

WHAT TO SUBMIT

1. Complete working API with all endpoints
2. Database schema and migration scripts
3. Comprehensive API documentation
4. Postman collection or API testing suite
5. Unit and integration tests with good coverage
6. Docker containerization and deployment scripts
7. Security audit and performance testing results
8. GitHub repository with detailed setup guide

HOW YOU WILL BE EVALUATED

1. Code Quality & Architecture (30%): Clean, maintainable code with proper structure
2. Functionality & Completeness (25%): All requirements implemented and working
3. Testing & Documentation (20%): Comprehensive tests and clear documentation
4. Performance & Optimization (15%): Efficient algorithms and optimized code
5. Best Practices & Standards (10%): Following industry conventions and patterns

SUBMISSION STEPS

1. Create a GitHub repository with your solution
2. Write a README.md with setup instructions
3. Add unit tests with good coverage
4. Deploy your app to a cloud platform
5. Email us your GitHub link and demo URL
6. Include screenshots or demo video

RECOMMENDED FOLDER STRUCTURE

```
project-name/  
% % % README.md  
% % % package.json  
% % % src/  
%   % % % components/  
%   % % % services/  
%   % % % index.js  
% % % tests/  
% % % docs/
```

TIPS

- Read all requirements before starting
- Plan your approach first
- Write clean, commented code
- Test your solution thoroughly
- Document your decisions
- Make sure it runs without errors

DEADLINE

Submit within 8-12 hours of receiving this assignment.

Questions? Email: technical-support@talentai.bid

Good luck! Ø=Þ€

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